AP PRECALCULUS

Unit 1 • Lesson 1.9 practice • Answer key

Lesson 1.9 • Rational Functions and Vertical Asymptotes

4 multiple-choice questions and 1 free-response question • Answer key with explanations

Question 1

Which line is a vertical asymptote of $r(x) = \dfrac{x+2}{x^2-x-6}$?

(A) $x = -2$

(B) $x = 3$

(C) $x = -2$ and $x = 3$

(D) $x = 6$

Solution

Answer: (B). $\dfrac{x+2}{(x-3)(x+2)}$: the factor $(x+2)$ cancels (a hole at $x = -2$), so only $x = 3$ is an asymptote.

Question 2

$\lim\limits_{x \to 1^+} \dfrac{2x}{x-1}$ is

(A) $+\infty$

(B) $-\infty$

(C) 2

(D) 0

Solution

Answer: (A). The numerator approaches $2 > 0$ and the denominator approaches 0 through positive values.

Question 3

$\lim\limits_{x \to -2^-} \dfrac{1}{(x+2)^2}$ is

(A) $-\infty$

(B) 0

(C) $\dfrac{1}{4}$

(D) $+\infty$

Solution

Answer: (D). The denominator is a square, positive on both sides of -2, and approaches 0.

Question 4

Which function has a vertical asymptote at $x = 4$?

(A) $\dfrac{x-4}{x+4}$

(B) $\dfrac{x^2-16}{x-4}$

(C) $\dfrac{x}{(x-4)^2}$

(D) $\dfrac{(x-4)^2}{x}$

Solution

Answer: (C). $x = 4$ is a zero of the denominator only. In (B) the factor cancels, giving a hole.

Question 5 (Free response)

Let $r(x) = \dfrac{x-1}{(x+2)(x-3)}$.

(a) Find the vertical asymptotes of the graph of r.
(b) For each vertical asymptote $x = a$, find $\lim\limits_{x \to a^-} r(x)$ and $\lim\limits_{x \to a^+} r(x)$. Show how you determine the signs.

Solution

(a) No factor cancels, so $x = -2$ and $x = 3$.
(b) Near $x = -2$ the numerator is close to -3 and $(x-3)$ is close to -5. For $x < -2$, $(x+2) < 0$, so the denominator is positive and $r < 0$: $\lim_{x \to -2^-} r(x) = -\infty$. For $x > -2$ the denominator is negative and $r > 0$: $\lim_{x \to -2^+} r(x) = +\infty$.
Near $x = 3$ the numerator is close to 2 and $(x+2)$ to 5. For $x < 3$ the denominator is negative: $\lim_{x \to 3^-} r(x) = -\infty$. For $x > 3$ it is positive: $\lim_{x \to 3^+} r(x) = +\infty$.